

Explore loops through puzzles, predict code, and build your own Scratch project.

NAME

First Last

DATE

__ / __ / __

TIER

A · B · C

🔗 PART 1 OF 5 · WARM-UP (NO SCRATCH!) · ~10 MIN

✓ ANSWER KEY

Think Like a Computer

🧩 PUZZLE 1 — BE THE COMPUTER (ALL TIERS)

Your teacher said: "Do 5 jumping jacks." Use a loop — don't repeat the action 5 times!

✓ With a loop — fill in the blanks:

○ **repeat** 5 move jump → Lines: 2

What does a loop save you from doing?

Writing the same instruction 5 separate times. A loop writes it once inside repeat 5 — saves 3 extra blocks!

🔗 PUZZLE 2 — REPEAT OR FOREVER? (ALL TIERS)

SITUATION	REPEAT N?	FOREVER?
Wave hello 3 times	✓	
Ball bounces while game runs		✓
Draw a triangle (3 sides)	✓	
Sprite dancing costume loop		✓

Key rule: repeat N = known count · forever = runs until stopped

📄 PUZZLE 3 — CODE SAVINGS (TIER B + C)

Square without loop = 8 lines. With ○ **repeat 4** = 3 lines. Saved: 5

Hexagon (6 sides): Without: 12 With: 3 Turn°: 60°

Formula: $360 \div \text{sides} = \text{turn angle}$

📖 PART 2 OF 5 · VOCABULARY CHECK

✓ ANSWER KEY

Loop Words

abc MATCH — WRITE THE LETTER (A–F)

A	An instruction that makes code run multiple times	loop
B	One single pass through a loop	iteration
C	Runs forever — stops only at Stop button	forever
D	Runs exactly N times, then stops	repeat N
E	A sprite's outfit — switching fast = animation	costume
F	How AI gets smarter — loops through examples	training

🗣️ IN MY OWN WORDS (FINISH EACH STARTER)

A loop is like...

...a merry-go-round — it keeps doing the same thing over and over until you tell it to stop.

I'd use ○ **forever** instead of ○ **repeat** when...

...I don't know how many times it should happen — like a bouncing ball or sprite checking for input.

A loop saves a programmer time because...

...you write the code once and the loop handles repetition automatically — no copy-pasting!

🔗 PART 3 OF 5 · PREDICT THE CODE

✓ ANSWER KEY

What Will Scratch Do?

📄 PREDICT 1 — SQUARE SCRIPT (ALL)

Script: ○ **repeat 4** → move 100 + turn 90°

Shape drawn: Square □

Loop runs: 4 times

Total degrees turned: 360°

Sketch it Perfect square — 4 equal sides, sprite returns to start. $4 \times 90^\circ = 360^\circ$

📄 PREDICT 2 — DANCING SCRIPT (B + C)

Script: ○ **forever** → next costume + wait 0.3s

Change 0.3s → 0.05s: what happens?

Animation becomes much faster — sprite flickers/dances rapidly. Eye perceives smooth motion.

Costumes in 10 sec: $10 \div 0.3 = \underline{\approx 33}$

Why can't you use ○ **repeat N** here?

📄 PREDICT 3 — NESTED LOOPS (TIER C)

Script: ○ **repeat 6** → [○ **repeat 4** → square] → turn 60°

Inner loop runs total: $6 \times 4 = \underline{24}$

Outer shape ($6 \times 60^\circ = 360^\circ$):

Sketch it Mandala / flower of squares — 6 squares arranged in a full 360° circle, 60° apart.

The dance must run the whole game — you don't know the exact count. Only forever loops indefinitely.

🔗 PART 4 OF 5 · OPEN BUILD · 30-40 MIN

✓ ANSWER KEY

Your Scratch Build — Choose Your Tier

A TIER A — DRAW A SQUARE

Use ☐ **repeat 4** to draw a square. Then change to 3 and 6.

Blocks used: 2 Changed repeat to 3: triangle

Must also change turn to 120° ($360 \div 3 = 120$). Keeping 90° leaves an open shape!

One thing I'm proud of:

(Accept any genuine reflection — e.g., "I figured out the turn formula myself!")

B TIER B — SQUARE + DANCE

Build both loops: ☐ **repeat 4** square & ☐ **forever** dance.

Script 1 plan (square): **When** ☐ → **repeat 4** → **[move 100, turn 90]**

Script 2 plan (dance): **When** ☐ → **forever** → **[next costume, wait 0.3]**

Lines saved: Without 8 · With 3 · Saved 5

C TIER C — NESTED FLOWER

Build the nested loop mandala. Fill in the table:

OUTER REPEAT	TURN ÷ 360	TOTAL INNER
4	90°	$4 \times 4 = 16$
6	60°	$6 \times 4 = 24$
8	45°	$8 \times 4 = 32$

As outer count increases:

More petals + smaller turn angle. Total inner work = outer × inner — mandala grows more complex!

🔗 PART 5 OF 5 · AI MOMENT

✓ ANSWER KEY

How Does AI Practice? It Loops!

🧠 THE CONNECTION: YOUR CODE = AI'S TRAINING ENGINE

When AI learns to recognize cat photos, it doesn't look at each photo once — it **loops** through thousands over and over, improving each time. A nested loop = AI going through every example for every training round. Total work = outer × inner.

If AI only looped through 3 cat photos, would it recognize cats well? Why?

No — it would memorize those 3 but fail on any new photo ("overfitting"). Like learning math by practicing only 3 problems — not enough variation to generalize to real-world inputs.

💬 THINK + WRITE

In Scratch, what's the AI equivalent of your ☐ **repeat 4** loop?

One training epoch — one complete pass through all training data. AI's repeat 10,000 = looks at examples 10,000 times, getting smarter each pass.

A real-life thing that loops to make you better:

Piano scales, push-up reps, reading flashcards — repetition builds skill, just like AI training!

Explain "total work = outer × inner" in your own words:

6 chapters × 4 pages each = 24 pages total. Outer loop × inner loop = total actions performed.

✓ EXIT TICKET — ANSWER KEY

1 One thing I learned about loops:

Loops save code. All closed shapes turn exactly 360°. Nested loops: total work = outer × inner.

2 When to use ☐ forever vs ☐ repeat N :

forever = unknown duration (game running). repeat N = known exact count (shape sides, training epochs).

3 One question I still have:

(Accept any genuine question — e.g., "Can loops have conditions?" or "How do I loop backwards?")